```
# Data handling
import pandas as pd
import numpy as np

# Data visualization
import matplotlib.pyplot as plt

# Database
import sqlite3

# Machine learning
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

# Model evaluation
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    confusion_matrix,
    ConfusionMatrixDisplay,
    classification_report,
    RocCurveDisplay
)

# Display settings
pd.set_option("display.max_columns", None)

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

```
dataset_url = (
    "https://raw.githubusercontent.com/IBM/"
    "telco-customer-churn-on-icp4d/master/data/"
    "Telco-Customer-Churn.csv"
)

df = pd.read_csv(dataset_url)

print("Dataset loaded successfully.")
```

```
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

```
Dataset loaded successfully.
Rows: 7043
Columns: 21
```

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneServi
0	7590-VHVEG	Female	0	Yes	No	1	N
1	5575-GNVDE	Male	0	No	No	34	Y
2	3668-QPYBK	Male	0	No	No	2	Y
3	7795-CFOCW	Male	0	No	No	45	N
4	9237-HQITU	Female	0	No	No	2	Y

1. Dataset Inspection

```
print("Dataset shape:", df.shape)

print("\nColumn names:")
print(df.columns.tolist())

print("\nDataset information:")
df.info()
```

```
Dataset shape: (7043, 21)

Column names:
['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
```

```
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

```
1 df.dtypes
```

	0
customerID	object
gender	object
SeniorCitizen	int64
Partner	object
Dependents	object
tenure	int64
PhoneService	object
MultipleLines	object
InternetService	object
OnlineSecurity	object
OnlineBackup	object
DeviceProtection	object
TechSupport	object
StreamingTV	object
StreamingMovies	object
Contract	object
PaperlessBilling	object
PaymentMethod	object
MonthlyCharges	float64
TotalCharges	object
Churn	object

dtype: object

```
missing_values = df.isnull().sum()

missing_values[missing_values > 0]
```

	0

dtype: int64

```
blank_total_charges = (df["TotalCharges"].str.strip() == "").sum()

print("Blank values in TotalCharges:", blank_total_charges)
```

```
Blank values in TotalCharges: 11
```

2. Data Cleaning

```
clean_df = df.copy()
```

```
clean_df["TotalCharges"] = pd.to_numeric(
    clean_df["TotalCharges"],
    errors="coerce"
)

print("Missing TotalCharges values:")
print(clean_df["TotalCharges"].isnull().sum())
```

```
Missing TotalCharges values:
11
```

```
clean_df = clean_df.dropna(subset=["TotalCharges"]).copy()

print("Shape after removing missing values:", clean_df.shape)
```

```
Shape after removing missing values: (7032, 21)
```

```
clean_df["ChurnFlag"] = clean_df["Churn"].map({
    "No": 0,
    "Yes": 1
})

clean_df[["Churn", "ChurnFlag"]].head()
```

	Churn	ChurnFlag
0	No	0
1	No	0
2	Yes	1
3	No	0
4	Yes	1

```
clean_df["SeniorCitizenLabel"] = clean_df["SeniorCitizen"].map({
    0: "No",
    1: "Yes"
})
```

```
clean_df["TenureGroup"] = pd.cut(
    clean_df["tenure"],
    bins=[-1, 12, 24, 48, 60, 72],
    labels=[
        "0-12 Months",
        "13-24 Months",
        "25-48 Months",
        "49-60 Months",
        "61-72 Months"
    ]
)
```

```
clean_df["MonthlyChargeGroup"] = pd.cut(
    clean_df["MonthlyCharges"],
    bins=[0, 35, 70, 100, float("inf")],
    labels=[
        "Low",
        "Medium",
        "High",
        "Very High"
    ],
    include_lowest=True
)
```

```
clean_df[
    [
        "customerID",
        "tenure",
        "TenureGroup",
        "MonthlyCharges",
        "MonthlyChargeGroup",
        "Churn"
    ]
].head()
```

	customerID	tenure	TenureGroup	MonthlyCharges	MonthlyChargeGroup	Churn
0	7590-VHVEG	1	0-12 Months	29.85	Low	No
1	5575-GNVDE	34	25-48 Months	56.95	Medium	No
2	3668-QPYBK	2	0-12 Months	53.85	Medium	Yes
3	7795-CFOCW	45	25-48 Months	42.30	Medium	No
4	9237-HQITU	2	0-12 Months	70.70	High	Yes

```
print("Duplicate rows:", clean_df.duplicated().sum())
print(
    "Duplicate customer IDs:",
    clean_df["customerID"].duplicated().sum()
)
```

```
Duplicate rows: 0
Duplicate customer IDs: 0
```


```
print("Duplicate rows:", clean_df.duplicated().sum())
print(
    "Duplicate customer IDs:",
    clean_df["customerID"].duplicated().sum()
)
```

```
Duplicate rows: 0
Duplicate customer IDs: 0
```


```
clean_df.describe().T
```

	count	mean	std	min	25%	50%	
SeniorCitizen	7032.0	0.162400	0.368844	0.00	0.0000	0.000	
tenure	7032.0	32.421786	24.545260	1.00	9.0000	29.000	5
MonthlyCharges	7032.0	64.798208	30.085974	18.25	35.5875	70.350	8
TotalCharges	7032.0	2283.300441	2266.771362	18.80	401.4500	1397.475	379
ChurnFlag	7032.0	0.265785	0.441782	0.00	0.0000	0.000	

```
clean_df.describe(include="object").T
```

	count	unique	top	freq
customerID	7032	7032	3186-AJIEK	1
gender	7032	2	Male	3549
Partner	7032	2	No	3639
Dependents	7032	2	No	4933
PhoneService	7032	2	Yes	6352
MultipleLines	7032	3	No	3385
InternetService	7032	3	Fiber optic	3096
OnlineSecurity	7032	3	No	3497
OnlineBackup	7032	3	No	3087
DeviceProtection	7032	3	No	3094
TechSupport	7032	3	No	3472
StreamingTV	7032	3	No	2809
StreamingMovies	7032	3	No	2781
Contract	7032	3	Month-to-month	3875
PaperlessBilling	7032	2	Yes	4168
PaymentMethod	7032	4	Electronic check	2365
Churn	7032	2	No	5163
SeniorCitizenLabel	7032	2	No	5890

3. Key Performance Indicators

```
total_customers = clean_df["customerID"].nunique()
churned_customers = clean_df["ChurnFlag"].sum()
active_customers = total_customers - churned_customers
churn_rate = clean_df["ChurnFlag"].mean() * 100

average_monthly_charge = clean_df["MonthlyCharges"].mean()
average_tenure = clean_df["tenure"].mean()
total_revenue = clean_df["TotalCharges"].sum()

print(f"Total Customers: {total_customers:,}")
print(f"Active Customers: {active_customers:,}")
print(f"Churned Customers: {churned_customers:,}")
print(f"Churn Rate: {churn_rate:.2f}%")
print(f"Average Monthly Charge: ${average_monthly_charge:,.2f}")
print(f"Average Tenure: {average_tenure:.2f} months")
print(f"Total Historical Charges: ${total_revenue:,.2f}")
```

```
Total Customers: 7,032
Active Customers: 5,163
Churned Customers: 1,869
Churn Rate: 26.58%
Average Monthly Charge: $64.80
Average Tenure: 32.42 months
Total Historical Charges: $16,056,168.70
```

```
kpi_data = {
    "KPI": [
        "Total Customers",
        "Active Customers",
        "Churned Customers",
        "Churn Rate",
        "Average Monthly Charge",
        "Average Tenure",
        "Total Historical Charges"
    ],
    "Value": [
        total_customers,
        active_customers,
        churned_customers,
        f"{churn_rate:.2f}%",
        f"${average_monthly_charge:,.2f}",
        f"{average_tenure:.2f} months",
        f"${total_revenue:,.2f}"
    ]
}

kpi_df = pd.DataFrame(kpi_data)

kpi_df
```

	KPI	Value
0	Total Customers	7032
1	Active Customers	5163
2	Churned Customers	1869
3	Churn Rate	26.58%
4	Average Monthly Charge	$64.80
5	Average Tenure	32.42 months
6	Total Historical Charges	$16,056,168.70

Next steps: Generate code with kpi_df | New interactive sheet

4. Exploratory Data Analysis

```
churn_counts = clean_df["Churn"].value_counts()

churn_counts
```

Churn	count
No	5163
Yes	1869

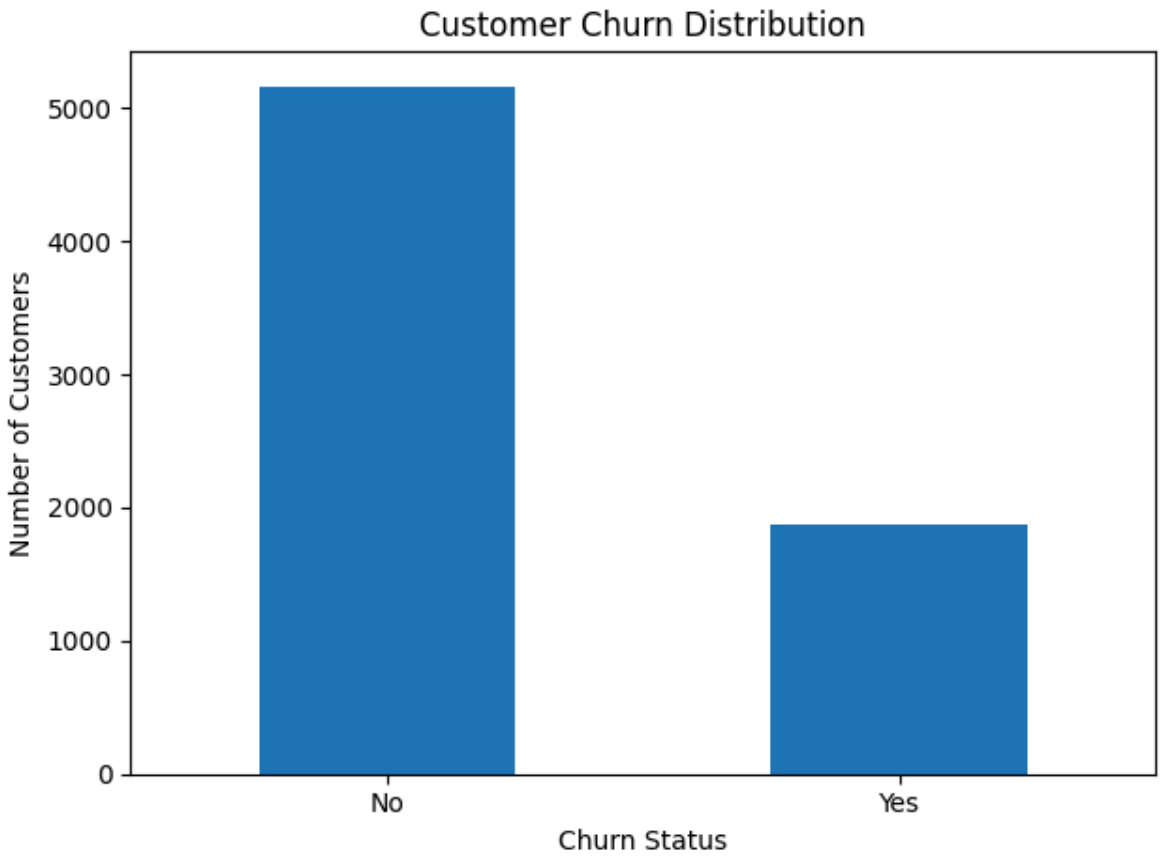
dtype: int64

```
churn_percentages = (
    clean_df["Churn"]
    .value_counts(normalize=True)
    .mul(100)
    .round(2)
)

churn_percentages
```

Churn	proportion
No	73.42
Yes	26.58

dtype: float64

```
churn_counts.plot(
    kind="bar",
    title="Customer Churn Distribution"
)

plt.xlabel("Churn Status")
plt.ylabel("Number of Customers")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

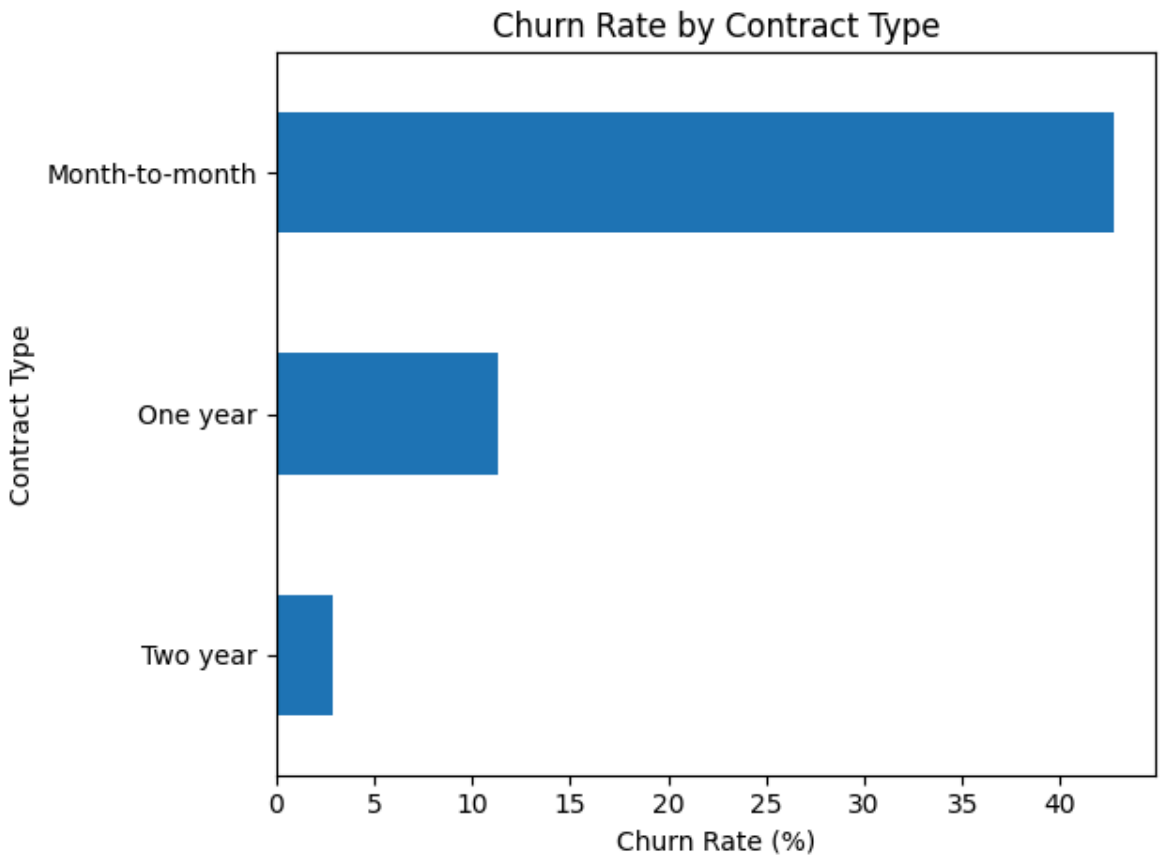



```
contract_analysis = pd.crosstab(
    clean_df["Contract"],
    clean_df["Churn"],
    normalize="index"
).mul(100).round(2)

contract_analysis
```

Churn	No	Yes
Contract		
Month-to-month	57.29	42.71
One year	88.72	11.28
Two year	97.15	2.85

Next steps: Generate code with `contract_analysis` New interactive sheet

```
contract_analysis["Yes"].sort_values().plot(
    kind="barh",
    title="Churn Rate by Contract Type"
)

plt.xlabel("Churn Rate (%)")
plt.ylabel("Contract Type")
plt.tight_layout()
plt.show()
```

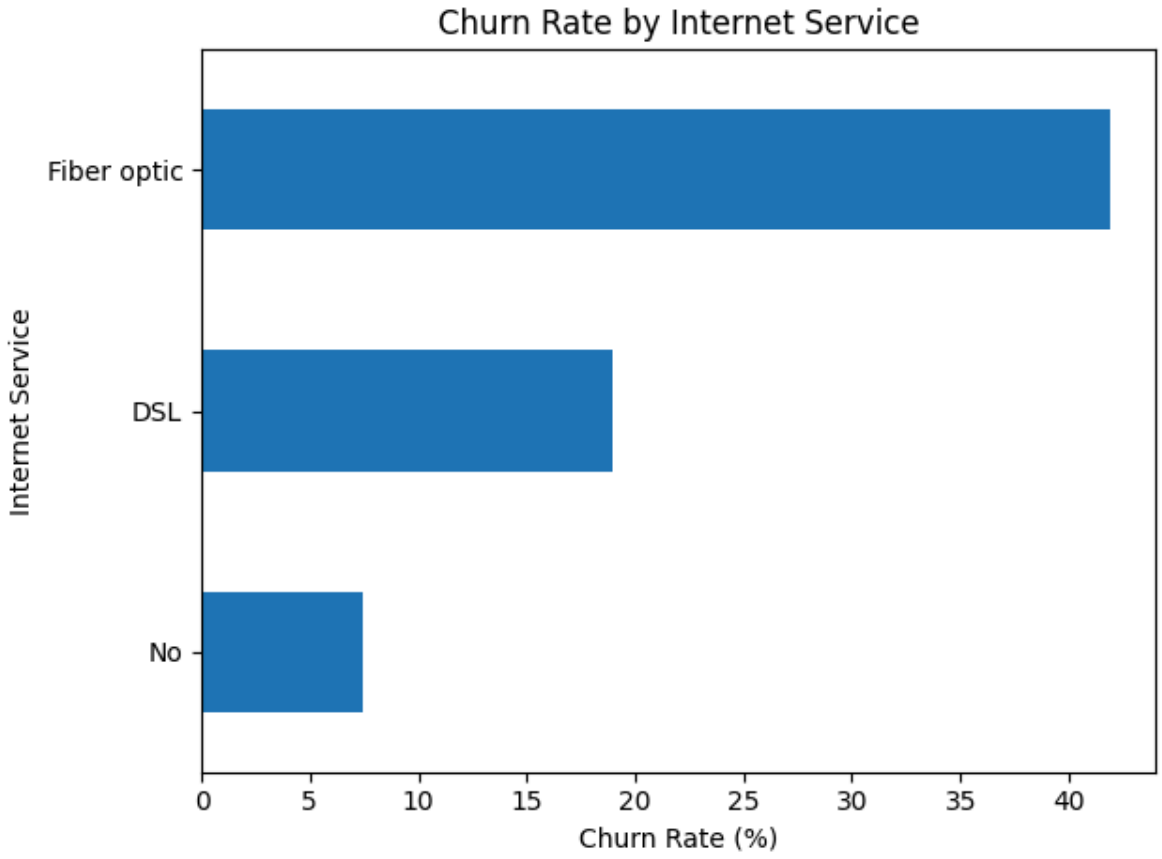



```
internet_analysis = pd.crosstab(
    clean_df["InternetService"],
    clean_df["Churn"],
    normalize="index"
).mul(100).round(2)

internet_analysis
```

InternetService	No	Yes
DSL	81.00	19.00
Fiber optic	58.11	41.89
No	92.57	7.43

Next steps: Generate code with `internet_analysis` New interactive sheet

```
internet_analysis["Yes"].sort_values().plot(
    kind="barh",
    title="Churn Rate by Internet Service"
)

plt.xlabel("Churn Rate (%)")
plt.ylabel("Internet Service")
plt.tight_layout()
plt.show()
```

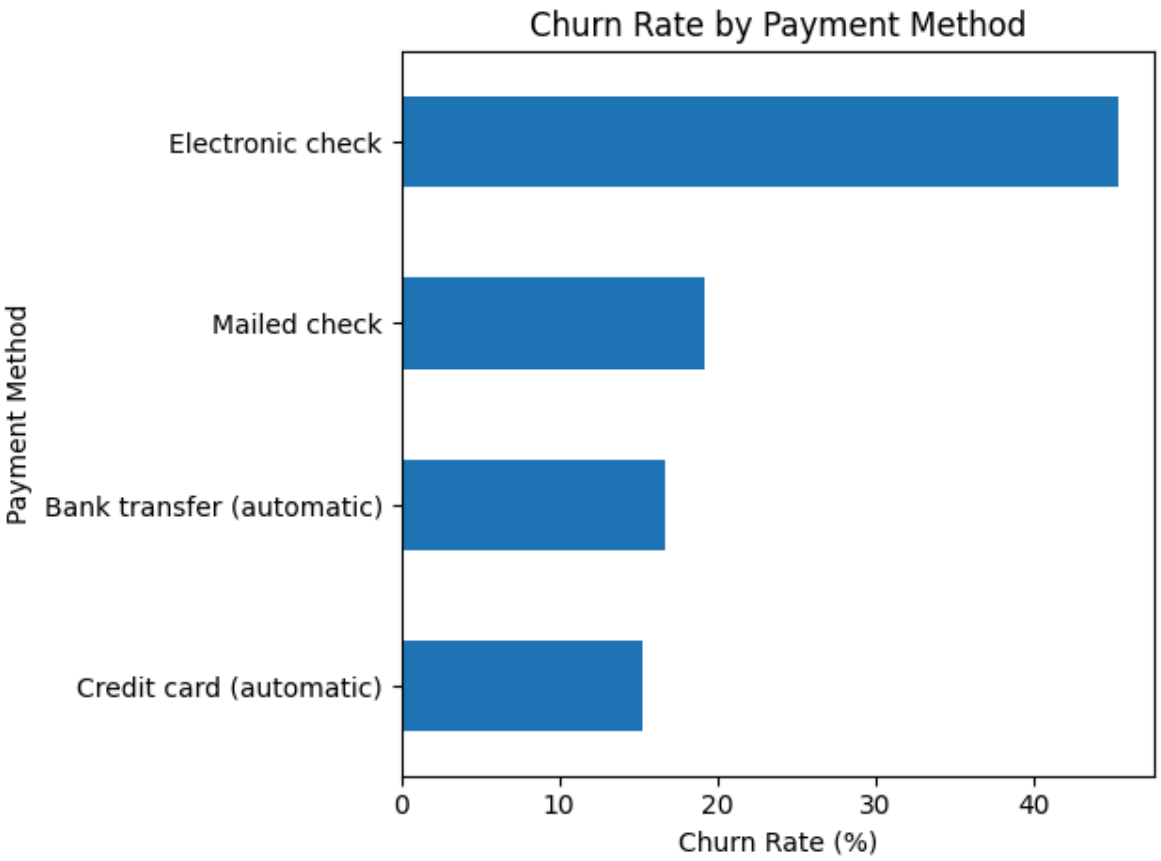



```
payment_analysis = pd.crosstab(
    clean_df["PaymentMethod"],
    clean_df["Churn"],
    normalize="index"
).mul(100).round(2)

payment_analysis
```

PaymentMethod	No	Yes
Bank transfer (automatic)	83.27	16.73
Credit card (automatic)	84.75	15.25
Electronic check	54.71	45.29
Mailed check	80.80	19.20

Next steps: Generate code with `payment_analysis` New interactive sheet

```
payment_analysis["Yes"].sort_values().plot(
    kind="barh",
    title="Churn Rate by Payment Method"
)

plt.xlabel("Churn Rate (%)")
plt.ylabel("Payment Method")
plt.tight_layout()
plt.show()
```

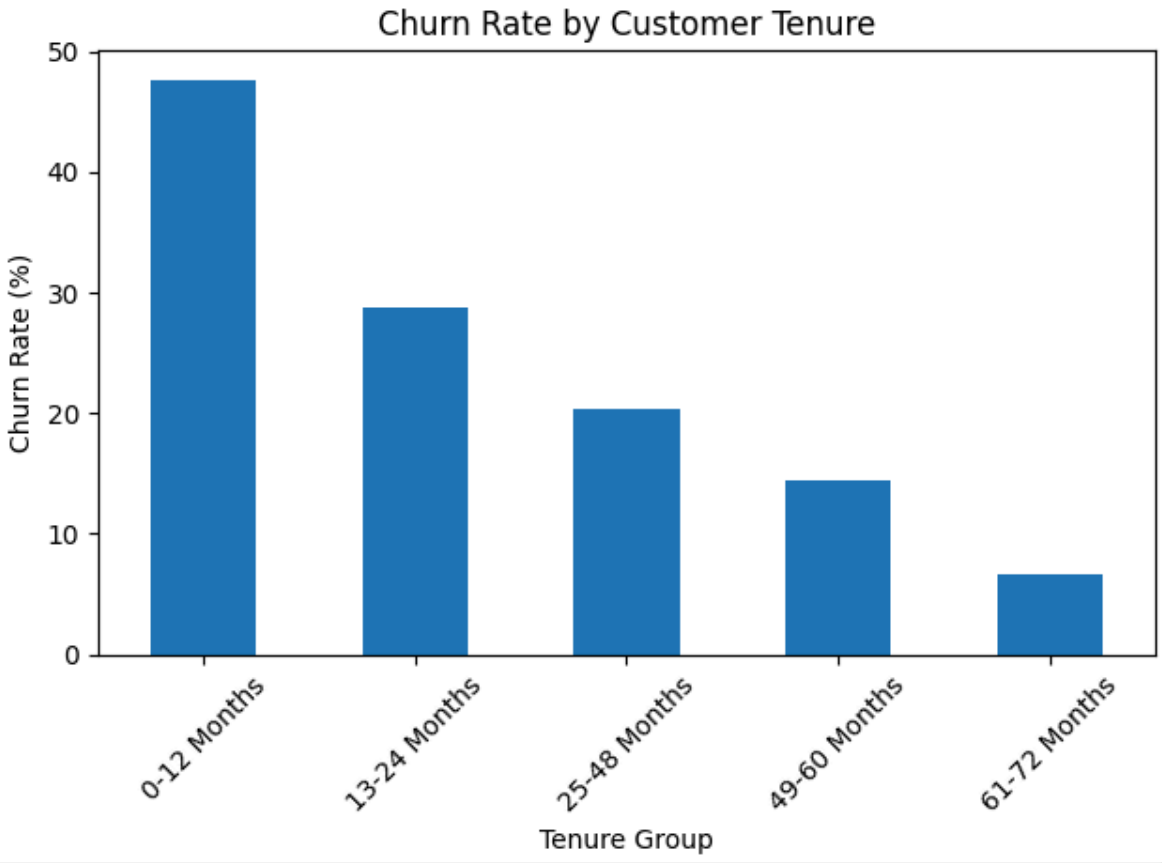



```
tenure_analysis = pd.crosstab(
    clean_df["TenureGroup"],
    clean_df["Churn"],
    normalize="index"
).mul(100).round(2)

tenure_analysis
```

Churn	No	Yes
TenureGroup		
0-12 Months	52.32	47.68
13-24 Months	71.29	28.71
25-48 Months	79.61	20.39
49-60 Months	85.58	14.42
61-72 Months	93.39	6.61

Next steps: Generate code with `tenure_analysis` New interactive sheet

```
tenure_analysis["Yes"].plot(
    kind="bar",
    title="Churn Rate by Customer Tenure"
)

plt.xlabel("Tenure Group")
plt.ylabel("Churn Rate (%)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



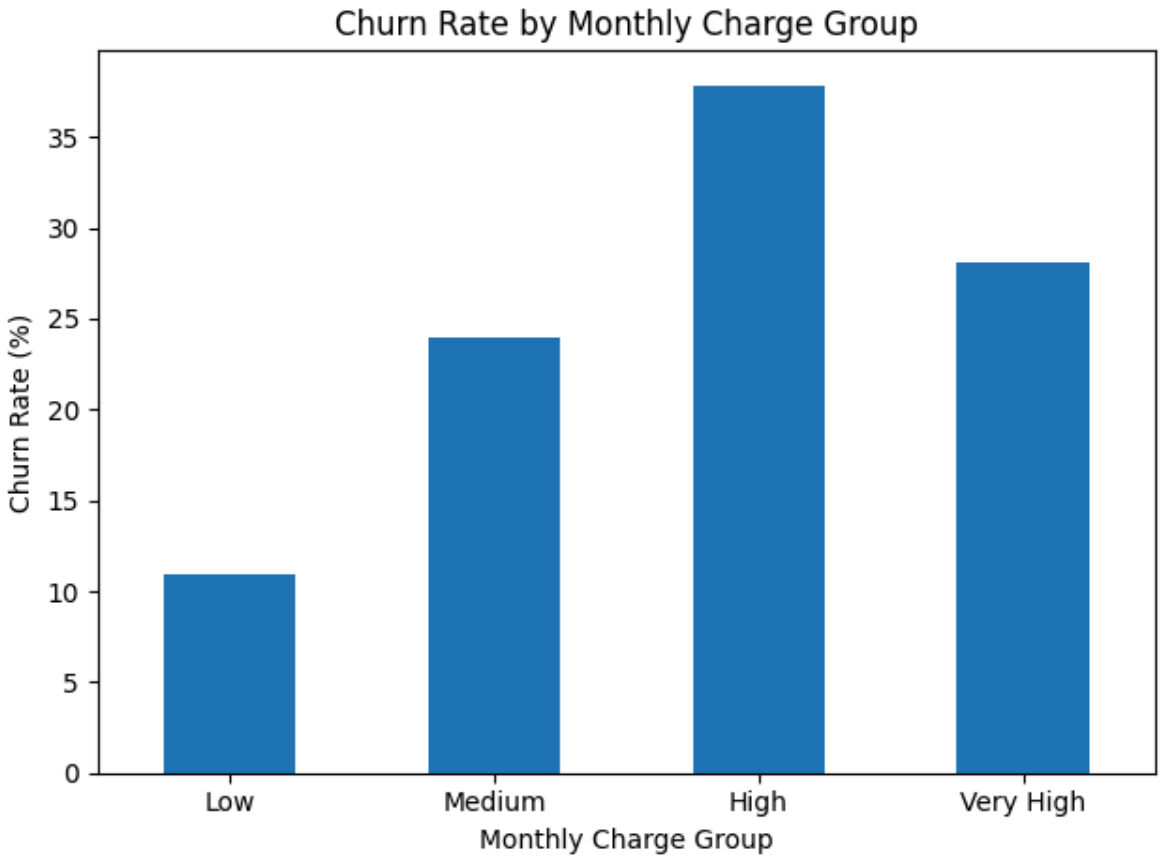

```
monthly_charge_analysis = pd.crosstab(
    clean_df["MonthlyChargeGroup"],
    clean_df["Churn"],
```

```
    normalize="index"
).mul(100).round(2)

monthly_charge_analysis
```

Churn	No	Yes
MonthlyChargeGroup		
Low	89.07	10.93
Medium	76.02	23.98
High	62.15	37.85
Very High	71.95	28.05

Next steps: Generate code with monthly_charge_analysis | New interactive sheet

```
monthly_charge_analysis["Yes"].plot(
    kind="bar",
    title="Churn Rate by Monthly Charge Group"
)

plt.xlabel("Monthly Charge Group")
plt.ylabel("Churn Rate (%)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
charge_comparison = clean_df.groupby("Churn")[
    ["MonthlyCharges", "TotalCharges", "tenure"]
].mean().round(2)

charge_comparison
```

Churn	MonthlyCharges	TotalCharges	tenure
No	61.31	2555.34	37.65
Yes	74.44	1531.80	17.98

Next steps: Generate code with `charge_comparison` New interactive sheet

5. SQL Customer Churn Analysis

```
connection = sqlite3.connect("customer_churn.db")

clean_df.to_sql(
    "customers",
    connection,
    if_exists="replace",
    index=False
)

print("Customer churn database created successfully.")
```

```
Customer churn database created successfully.
```

```
query = """
SELECT *
FROM customers
LIMIT 5;
"""

pd.read_sql_query(query, connection)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneServi
0	7590-VHVEG	Female	0	Yes	No	1	N
1	5575-GNVDE	Male	0	No	No	34	Y
2	3668-QPYBK	Male	0	No	No	2	Y
3	7795-CFOCW	Male	0	No	No	45	N
4	9237-HQITU	Female	0	No	No	2	Y

```
query = """
SELECT
    COUNT(*) AS total_customers,
    SUM(ChurnFlag) AS churned_customers,
    ROUND(
        100.0 * SUM(ChurnFlag) / COUNT(*),
        2
    ) AS churn_rate
FROM customers;
"""

pd.read_sql_query(query, connection)
```

	total_customers	churned_customers	churn_rate
0	7032	1869	26.58

```
query = """
SELECT
    Contract,
    COUNT(*) AS total_customers,
    SUM(ChurnFlag) AS churned_customers,
    ROUND(
        100.0 * SUM(ChurnFlag) / COUNT(*),
        2
    ) AS churn_rate
FROM customers
GROUP BY Contract
ORDER BY churn_rate DESC;
"""

pd.read_sql_query(query, connection)
```

	Contract	total_customers	churned_customers	churn_rate
0	Month-to-month	3875	1655	42.71
1	One year	1472	166	11.28
2	Two year	1685	48	2.85

```
query = """
SELECT
    PaymentMethod,
    COUNT(*) AS total_customers,
    SUM(ChurnFlag) AS churned_customers,
    ROUND(
        100.0 * SUM(ChurnFlag) / COUNT(*),
        2
    ) AS churn_rate
FROM customers
GROUP BY PaymentMethod
ORDER BY churn_rate DESC;
"""

pd.read_sql_query(query, connection)
```

	PaymentMethod	total_customers	churned_customers	churn_rate
0	Electronic check	2365	1071	45.29
1	Mailed check	1604	308	19.20
2	Bank transfer (automatic)	1542	258	16.73
3	Credit card (automatic)	1521	232	15.25

```
query = """
SELECT
    customerID,
    Contract,
    tenure,
    MonthlyCharges,
    TotalCharges,
    PaymentMethod
FROM customers
WHERE ChurnFlag = 1
  AND TotalCharges > (
      SELECT AVG(TotalCharges)
      FROM customers
  )
ORDER BY TotalCharges DESC
LIMIT 20;
"""
```

```
18
19 pd.read_sql_query(query, connection)
```

	customerID	Contract	tenure	MonthlyCharges	TotalCharges	PaymentMetho
0	2889-FPWRM	One year	72	117.80	8684.80	Bank transf (automati
1	0201-OAMXR	One year	70	115.55	8127.60	Credit ca (automati
2	3886-CERTZ	One year	72	109.25	8109.80	Electron che
3	1444-VVSGW	One year	70	115.65	7968.85	Credit ca (automati
4	5271-YNWVR	Two year	68	113.15	7856.00	Electron che
5	8199-ZLLSA	One year	67	118.35	7804.15	Bank transf (automati
6	9053-JZFKV	Two year	67	116.20	7752.30	Credit ca (automati
7	1555-DJEQW	Two year	70	114.20	7723.90	Bank transf (automati
8	3259-FDWOY	Two year	71	106.00	7723.70	Bank transf (automati
9	7317-GGVPB	Two year	71	108.60	7690.90	Credit ca (automati
10	0917-EZOLA	Two year	72	104.15	7689.95	Bank transf (automati
11	1984-FCOWB	One year	70	109.50	7674.55	Electron che
12	3192-NQECA	Two year	68	110.00	7611.85	Bank transf (automati
13	5287-QWLKY	Month-to-month	71	105.10	7548.10	Credit ca (automati
14	0748-RDGGM	One year	70	109.50	7534.65	Bank transf (automati
15	2834-JRTUA	Two year	71	108.05	7532.15	Electron che
16	9835-ZIITK	One year	66	110.85	7491.75	Electron che
17	2659-VXMWZ	One year	67	111.30	7482.10	Electron che
18	5502-RLUYV	Month-to-month	69	103.95	7446.90	Electron che
19	7632-MNYOY	One year	66	110.90	7432.05	Credit ca (automati

```
query = """
SELECT
    customerID,
    Contract,
    tenure,
    MonthlyCharges,
    TotalCharges,
    Churn,
    CASE
        WHEN Contract = 'Month-to-month'
             AND tenure <= 12
             AND MonthlyCharges >= 70
        THEN 'High Risk'

        WHEN Contract = 'Month-to-month'
             OR tenure <= 24
        THEN 'Medium Risk'

        ELSE 'Low Risk'
    END AS RiskSegment
FROM customers
ORDER BY TotalCharges DESC;
"""

risk_segments_df = pd.read_sql_query(query, connection)

risk_segments_df.head()
```

	customerID	Contract	tenure	MonthlyCharges	TotalCharges	Churn	RiskSeg
0	2889-FPWRM	One year	72	117.80	8684.80	Yes	Low
1	7569-NMZYQ	Two year	72	118.75	8672.45	No	Low
2	9739-JLPQJ	Two year	72	117.50	8670.10	No	Low
3	9788-HNGUT	Two year	72	116.95	8594.40	No	Low
4	8879-XUAHX	Two year	71	116.25	8564.75	No	Low

Next steps: Generate code with `risk_segments_df` New interactive sheet

6. Machine Learning Model Development

```
model_df = clean_df.drop(
    columns=[
        "customerID",
        "Churn",
        "ChurnFlag",
        "TenureGroup",
        "MonthlyChargeGroup",
        "SeniorCitizenLabel"
    ]
).copy()
```

```
X = model_df
y = clean_df["ChurnFlag"]

print("Feature shape:", X.shape)
print("Target shape:", y.shape)
```

```
Feature shape: (7032, 19)
Target shape: (7032,)
```

```
numerical_columns = X.select_dtypes(
    include=["int64", "float64"]
).columns.tolist()

categorical_columns = X.select_dtypes(
    include=["object", "category"]
).columns.tolist()

print("Numerical columns:")
print(numerical_columns)

print("\nCategorical columns:")
print(categorical_columns)
```

```
Numerical columns:
['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges']

Categorical columns:
['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'Interne
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training records:", X_train.shape[0])
print("Testing records:", X_test.shape[0])
```

```
Training records: 5625
Testing records: 1407
```

```
numerical_transformer = Pipeline(
    steps=[
        ("scaler", StandardScaler())
    ]
)

categorical_transformer = Pipeline(
    steps=[
        (
            "encoder",
            OneHotEncoder(
                handle_unknown="ignore",
                sparse_output=False
            )
        )
    ]
)

preprocessor = ColumnTransformer(
    transformers=[
        (
            "numerical",
            numerical_transformer,
            numerical_columns
        ),
        (
            "categorical",
            categorical_transformer,
            categorical_columns
        )
    ]
)
```

```
OneHotEncoder(
    handle_unknown="ignore",
    sparse_output=False
)
```

▾ OneHotEncoder

```
OneHotEncoder(handle_unknown='ignore', sparse_output=False)
```

```
logistic_pipeline = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        (
            "model",
```

```
            LogisticRegression(
                max_iter=1000,
                class_weight="balanced",
                random_state=42
            )
        )
    ]
)

logistic_pipeline.fit(X_train, y_train)

logistic_predictions = logistic_pipeline.predict(X_test)
logistic_probabilities = logistic_pipeline.predict_proba(X_test)[:

print("Logistic Regression trained successfully.")
```

```
Logistic Regression trained successfully.
```

```
decision_tree_pipeline = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        (
            "model",
            DecisionTreeClassifier(
                max_depth=5,
                min_samples_split=20,
                class_weight="balanced",
                random_state=42
            )
        )
    ]
)

decision_tree_pipeline.fit(X_train, y_train)

decision_tree_predictions = decision_tree_pipeline.predict(X_test)
decision_tree_probabilities = (
    decision_tree_pipeline.predict_proba(X_test)[:, 1]
)

print("Decision Tree trained successfully.")
```

```
Decision Tree trained successfully.
```

```
random_forest_pipeline = Pipeline(
    steps=[
        ("preprocessor", preprocessor),
        (
            "model",
            RandomForestClassifier(
```

```
                n_estimators=200,
                max_depth=10,
                min_samples_split=10,
                class_weight="balanced",
                random_state=42,
                n_jobs=-1
            )
        )
    ]
)

random_forest_pipeline.fit(X_train, y_train)

random_forest_predictions = random_forest_pipeline.predict(X_test)
random_forest_probabilities = (
    random_forest_pipeline.predict_proba(X_test)[:, 1]
)

print("Random Forest trained successfully.")
```

```
Random Forest trained successfully.
```

```
def evaluate_model(
    model_name,
    actual,
    predictions,
    probabilities
):
    return {
        "Model": model_name,
        "Accuracy": accuracy_score(actual, predictions),
        "Precision": precision_score(actual, predictions),
        "Recall": recall_score(actual, predictions),
        "F1 Score": f1_score(actual, predictions),
        "ROC-AUC": roc_auc_score(actual, probabilities)
    }
```

```
model_results = []

model_results.append(
    evaluate_model(
        "Logistic Regression",
        y_test,
        logistic_predictions,
        logistic_probabilities
    )
)

model_results.append(
    evaluate_model(
        "Decision Tree",
```

```
        y_test,
        decision_tree_predictions,
        decision_tree_probabilities
    )
)

model_results.append(
    evaluate_model(
        "Random Forest",
        y_test,
        random_forest_predictions,
        random_forest_probabilities
    )
)

results_df = pd.DataFrame(model_results)

results_df = results_df.sort_values(
    by="ROC-AUC",
    ascending=False
).reset_index(drop=True)

results_df.round(4)
```

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.7257	0.4901	0.7968	0.6069	0.8351
1	Random Forest	0.7598	0.5349	0.7380	0.6202	0.8302
2	Decision Tree	0.7335	0.4992	0.7941	0.6130	0.8276

```
print(
    classification_report(
        y_test,
        logistic_predictions,
        target_names=["Stayed", "Churned"]
    )
)
```

```
              precision    recall  f1-score   support

      Stayed       0.90      0.70      0.79      1033
     Churned       0.49      0.80      0.61       374

    accuracy                           0.73      1407
   macro avg       0.70      0.75      0.70      1407
weighted avg       0.79      0.73      0.74      1407
```

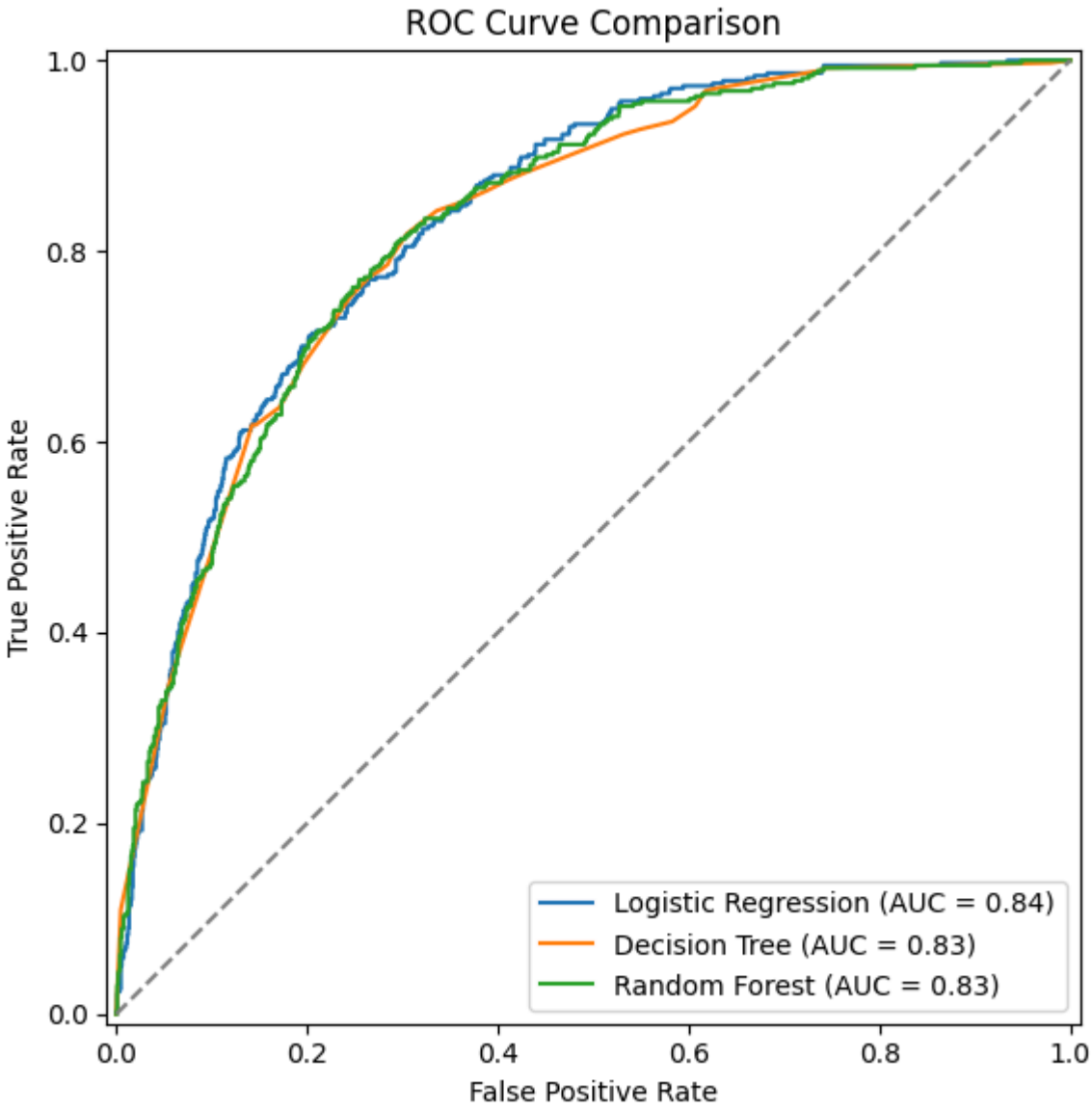
```
print(
    classification_report(
        y_test,
```

```
        random_forest_predictions,
        target_names=["Stayed", "Churned"]
    )
```

```
              precision    recall  f1-score   support

      Stayed       0.89      0.77      0.82      1033
     Churned       0.53      0.74      0.62       374

    accuracy                           0.76      1407
   macro avg       0.71      0.75      0.72      1407
weighted avg       0.80      0.76      0.77      1407
```

```
from sklearn.metrics import RocCurveDisplay
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(8, 6))

RocCurveDisplay.from_predictions(
    y_test,
    logistic_probabilities,
    name="Logistic Regression",
    ax=ax
)

RocCurveDisplay.from_predictions(
    y_test,
    decision_tree_probabilities,
    name="Decision Tree",
    ax=ax
)

RocCurveDisplay.from_predictions(
    y_test,
    random_forest_probabilities,
    name="Random Forest",
    ax=ax
)

ax.plot([0, 1], [0, 1], linestyle="--", color="gray")
ax.set_title("ROC Curve Comparison")
ax.set_xlabel("False Positive Rate")
ax.set_ylabel("True Positive Rate")

plt.tight_layout()

plt.savefig(
    "roc_curve_comparison.png",
    dpi=300,
    bbox_inches="tight"
)
```

```
plt.show()

print("ROC curve saved successfully.")
```




```
ROC curve saved successfully.
```

```
feature_names = (
    random_forest_pipeline
    .named_steps["preprocessor"]
    .get_feature_names_out()
)

importance_values = (
    random_forest_pipeline
    .named_steps["model"]
    .feature_importances_
)

feature_importance_df = pd.DataFrame({
    "Feature": feature_names,
    "Importance": importance_values
})
```

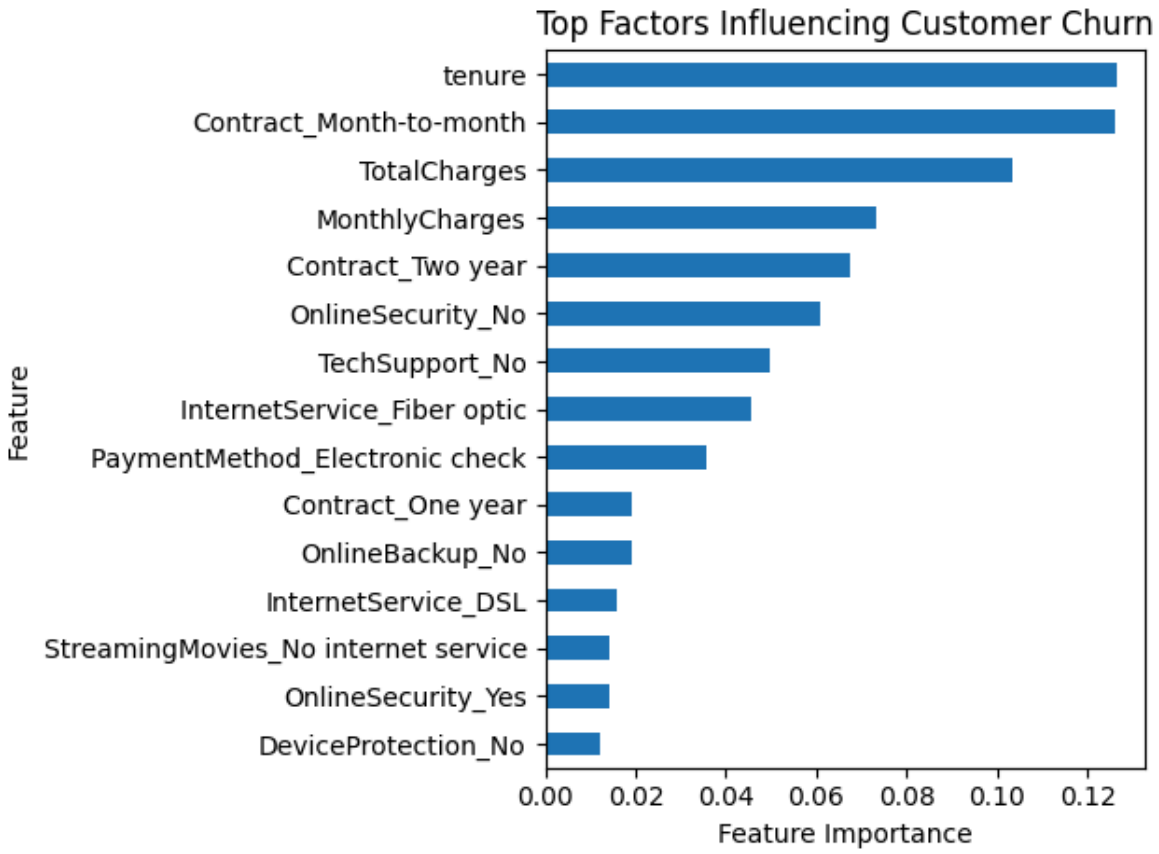
```
17
18 feature_importance_df["Feature"] = (
19     feature_importance_df["Feature"]
20     .str.replace("numerical__", "", regex=False)
21     .str.replace("categorical__", "", regex=False)
22 )
23
24 feature_importance_df = feature_importance_df.sort_values(
25     by="Importance",
26     ascending=False
27 ).reset_index(drop=True)
28
```

	Feature	Importance
0	tenure	0.126334
1	Contract_Month-to-month	0.126228
2	TotalCharges	0.103537
3	MonthlyCharges	0.073437
4	Contract_Two year	0.067382
5	OnlineSecurity_No	0.060995
6	TechSupport_No	0.049865
7	InternetService_Fiber optic	0.045553
8	PaymentMethod_Electronic check	0.035620
9	Contract_One year	0.019316
10	OnlineBackup_No	0.019016
11	InternetService_DSL	0.015906
12	StreamingMovies_No internet service	0.014145
13	OnlineSecurity_Yes	0.014133
14	DeviceProtection_No	0.012102

Next steps: Generate code with `feature_importance_df` New interactive sheet

```
1 top_features = feature_importance_df.head(15).sort_values(
2     by="Importance"
3 )
4
5 top_features.plot(
6     x="Feature",
7     y="Importance",
8     kind="barh",
9     legend=False,
10    title="Top Factors Influencing Customer Churn"
```

```
)

plt.xlabel("Feature Importance")
plt.ylabel("Feature")
plt.tight_layout()
plt.savefig(
    "feature_importance.png",
    dpi=300,
    bbox_inches="tight"
)
plt.show()
```




```
customer_features = clean_df[X.columns]

clean_df["PredictedChurn"] = logistic_pipeline.predict(
    customer_features
)

clean_df["ChurnProbability"] = logistic_pipeline.predict_proba(
    customer_features
)[:, 1]
```

```
clean_df["RiskLevel"] = pd.cut(
    clean_df["ChurnProbability"],
    bins=[0, 0.30, 0.60, 1.00],
    labels=[
        "Low Risk",
        "Medium Risk",
```

```
        "High Risk"
    ],
    include_lowest=True
)
```

```
clean_df[
    [
        "customerID",
        "Churn",
        "PredictedChurn",
        "ChurnProbability",
        "RiskLevel"
    ]
].head(10)
```

	customerID	Churn	PredictedChurn	ChurnProbability	RiskLevel
0	7590-VHVEG	No	1	0.809656	High Risk
1	5575-GNVDE	No	0	0.116409	Low Risk
2	3668-QPYBK	Yes	1	0.515596	Medium Risk
3	7795-CFOCW	No	0	0.081040	Low Risk
4	9237-HQITU	Yes	1	0.852570	High Risk
5	9305-CDSKC	Yes	1	0.911660	High Risk
6	1452-KIOVK	No	1	0.723692	High Risk
7	6713-OKOMC	No	1	0.564519	Medium Risk
8	7892-POOKP	Yes	1	0.814708	High Risk
9	6388-TABGU	No	0	0.029970	Low Risk

```
def recommend_action(row):
    if row["RiskLevel"] == "High Risk":
        if row["Contract"] == "Month-to-month":
            return "Offer discounted annual contract"

        if row["TechSupport"] == "No":
            return "Offer free technical support trial"

        return "Immediate retention outreach"

    if row["RiskLevel"] == "Medium Risk":
        return "Send personalized loyalty offer"

    return "Maintain regular engagement"

```

```

clean_df["RecommendedAction"] = clean_df.apply(
    recommend_action,
    axis=1
)
```

```
clean_df[
    [
```